

# ARIA: Adaptive Recurrent Intelligence Architecture

Morphogenic Attention and Slow-Pathway Consolidation for Continual Learning

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## Abstract

Catastrophic forgetting remains a fundamental obstacle in sequential learning. Most existing mitigations — Elastic Weight Consolidation (EWC), replay buffers, progressive networks — protect weights without changing how the model allocates capacity across tasks. We introduce ARIA, an architecture in which the capacity allocation *is* the learning signal. ARIA combines four jointly-trained mechanisms: (1) **Morphogenic Attention (MA)**, where heads split or merge based on learned viability scores; (2) **Plasticity-Gated MLP (PG-MLP)**, a dual fast/slow pathway whose gate routes each token toward volatile or consolidated representations; (3) **Architecture Genome Vector (AGV)**, a global latent vector that conditions all blocks via FiLM affine transforms; and (4) **Cognitive Budget Allocator (CBA)**, which predicts per-layer compute budgets from input complexity statistics. We additionally introduce **Slow-Pathway Consolidation (SPC)**, an EWC-style Fisher regulariser applied *only* to slow-pathway weights, allowing fast pathways to adapt freely while protecting consolidated knowledge. We further introduce three lightweight enhancements — Slow-Pathway Activation Distillation (SPAD), task-conditioned gate routing, and asymmetric learning rates — that together constitute ARIA-v2. On Split-MNIST (5 seeds, parameter-matched baselines), ARIA+SPC achieves **98.50%** average accuracy with **0.86%** forgetting, outperforming EWC (97.35%, 2.29%) by +1.15 points accuracy and reducing forgetting by **62%** relative, while requiring 50% fewer Fisher parameters than EWC.

## 1 Introduction

A learning system that encounters tasks sequentially faces a dilemma: plasticity (rapid adaptation to new tasks) and stability (retention of old tasks) are in direct tension [McCloskey and Cohen, 1989, French,

1999]. Neural networks trained by stochastic gradient descent exhibit extreme plasticity at the cost of stability — a phenomenon known as *catastrophic forgetting*.

The continual learning literature has addressed this through three broad strategies: regularisation (EWC [Kirkpatrick et al., 2017], SI [Zenke et al., 2017]), replay (DER++ [Buzzega et al., 2020], ER [Rolnick et al., 2019]), and progressive architectures (PNN [Rusu et al., 2016], PackNet [Mallya and Lazebnik, 2018]). Each strategy has a known failure mode: regularisation is too conservative for long task sequences, replay has memory cost, and progressive networks do not transfer across tasks.

ARIA takes a fourth approach: the architecture adapts its *structure* in response to task demands, distributing capacity dynamically so that both new and old knowledge can coexist without a fixed trade-off. The core insight is that the fast/slow dichotomy observed in biological memory [McClelland et al., 1995] can be implemented as a learned gate per neuron, with slow weights receiving reduced gradient updates during high-plasticity phases and stronger Fisher-based protection after consolidation.

### Contributions.

1. **Morphogenic Attention:** the first attention mechanism where head count is a learned, dynamic quantity, with principled split (specialisation) and merge (consolidation) operations governed by viability scores and cosine similarity.
2. **Plasticity-Gated MLP:** a dual-pathway MLP with a per-token learned gate and bimodal specialisation loss, inspired by complementary learning systems.
3. **Slow-Pathway Consolidation (SPC):** targeted Fisher regularisation over slow-pathway weights only, using 50% fewer Fisher parameters than EWC while improving specificity.

4. **Architecture Genome Vector:** a shared latent vector that decodes to skip probabilities, attention temperature, and FiLM scale/shift, conditioning all blocks with a single differentiable structural descriptor.
5. **Cognitive Budget Allocator:** a lightweight network predicting per-layer compute allocation from raw input statistics.

## 2 Related Work

**Regularisation methods.** EWC [Kirkpatrick et al., 2017] penalises movement of weights that were important for past tasks, measured by the diagonal Fisher information matrix. SI [Zenke et al., 2017] accumulates importance online during training. Both apply uniform regularisation across all parameters — including weights designed to be volatile, which conflicts with plasticity-gated architectures like ARIA.

**Replay methods.** DER++ [Buzzega et al., 2020] stores (input, logit) pairs and penalises divergence from stored predictions. ER [Rolnick et al., 2019] stores raw examples. Replay methods require a memory buffer whose size scales with the number of tasks. In our experiments, DER++ with a small buffer (200 samples) underperforms EWC significantly.

**Architecture methods.** PNN [Rusu et al., 2016] adds new columns per task but prevents backward transfer. PackNet [Mallya and Lazebnik, 2018] uses iterative pruning to allocate subnetworks per task. HAT [Serra et al., 2018] learns binary task masks. None dynamically restructure *within* an attention head.

**Dynamic architectures.** DEN [Yoon et al., 2018] expands hidden units based on a drift threshold — closest to ARIA in spirit, but operates at the unit level without attention or plasticity gating.

**Novelty-triggered Capacity Growth (NCG).** NCG [Poudel, 2025] is a prior architecture by the same author that addresses catastrophic forgetting via reactive capacity expansion: a 2-layer MLP with expandable hidden units adds 64 neurons when novelty is low and accuracy plateaus, regulated by online Lagrangian meta-parameters  $\alpha$ ,  $\beta$ ,  $\lambda$ . On Split-MNIST, NCG achieves 55.1% average accuracy, falling below EWC (73.2%), because growth is triggered too infrequently on short task sequences and the base model has limited representational depth. ARIA draws a

different conclusion from NCG’s failure: instead of asking *when* to grow capacity, ARIA asks *how* to structure learning so that growth is unnecessary — plasticity and stability are balanced inside the architecture itself through PG-MLP gating and targeted SPC regularisation.

**Complementary learning systems.** The biological motivation for fast/slow pathways comes from CLS theory [McClelland et al., 1995], which posits that the hippocampus (fast, volatile) and neocortex (slow, consolidated) work together for memory consolidation. ARIA’s PG-MLP is the first to implement this as a per-token learned gate with gradient dampening.

## 3 Method

### 3.1 Problem Setup

We consider the task-incremental continual learning setting. The model observes a sequence of tasks  $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_K$ , each with a dataset  $\mathcal{D}_t = \{(x_i^{(t)}, y_i^{(t)})\}$ . At test time, the task identity is provided (task-ID setting).

### 3.2 Architecture Overview

ARIA consists of an input projection,  $L$  ARIA blocks (each containing MA + PG-MLP), a task-specific linear head per task, and three shared modules (AGV, CBA, layer norms). Figure 1 illustrates the overall architecture.

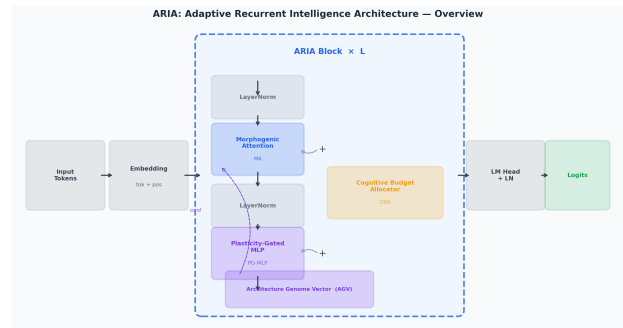


Figure 1: ARIA architecture. Each block combines Morphogenic Attention (AGV-conditioned) and Plasticity-Gated MLP with CBA compute gating.

For each block  $\ell$ :

$$h_\ell = b_\ell \cdot (z_\ell + \text{PG-MLP}(\text{LN}(z_\ell))) + (1 - b_\ell) \cdot h_{\ell-1} \quad (1)$$

where  $z_\ell = \text{MA}(\text{LN}(h_{\ell-1})) + h_{\ell-1}$  and  $b_\ell \in [0, 1]$  is the CBA-predicted budget for layer  $\ell$ .

### 3.3 Morphogenic Attention

Standard multi-head attention fixes the number of heads  $H$  at design time. Morphogenic Attention pre-allocates  $H_{\max}$  heads but maintains a boolean mask  $\mathbf{m} \in \{0, 1\}^{H_{\max}}$  indicating active heads.

Each head  $i$  has a scalar viability  $v_i \in \mathbb{R}$ , updated by backprop. The output of MA is:

$$\text{MA}(x) = \gamma \odot \left( \sum_{i \in \mathcal{A}} w_i \cdot \text{head}_i(x) \right) + \beta \quad (2)$$

where  $\mathcal{A} = \{i : m_i = 1\}$ ,  $w_i$  are softmax-normalised viabilities scaled by  $|\mathcal{A}|$ , and  $(\gamma, \beta)$  are FiLM parameters from the AGV.

**Split.** Every  $\Delta_{\text{morph}}$  steps, for each active head  $i$ : if  $\sigma(v_i) > \tau_{\text{split}}$ ,  $H_{\text{active}} < H_{\max}$ , and cooldown  $c_{\text{cool}}$  is satisfied, a new head  $j$  is created:

$$W^{(j)} \leftarrow W^{(i)} + \epsilon \cdot \mathcal{N}(0, I), \quad v_j \leftarrow v_i - 0.5 \quad (3)$$

**Merge.** For adjacent heads  $(i, j)$  with  $\cos(W_Q^{(i)}, W_Q^{(j)}) > \tau_{\text{merge}}$ ,  $H_{\text{active}} > 2$ , and cooldown satisfied:

$$W^{(i)} \leftarrow \frac{1}{2}(W^{(i)} + W^{(j)}), \quad m_j \leftarrow 0 \quad (4)$$

### 3.4 Plasticity-Gated MLP

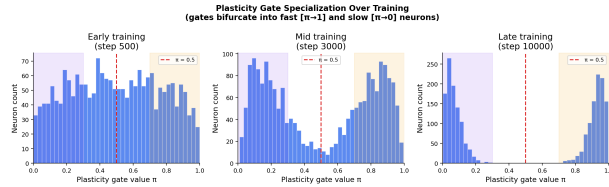


Figure 2: Plasticity gate  $\pi$  trajectories across tasks. High  $\pi$  (warm colours) indicates fast-pathway dominance; low  $\pi$  (cool colours) indicates consolidation.

The PG-MLP replaces the standard feed-forward layer with dual pathways:

$$\pi(x) = \sigma(\text{MLP}_{\text{gate}}(x)) \in (0, 1) \quad (5)$$

$$\text{PG-MLP}(x) = \pi(x) \cdot W_f^{(2)} \text{GeLU}(W_f^{(1)} x) + (1 - \pi(x)) \cdot W_s^{(2)} \text{GeLU}(W_s^{(1)} x) \quad (6)$$

**Bimodal specialisation loss.**

$$\mathcal{L}_{\text{plastic}} = \frac{\lambda_{\pi}}{\pi(1 - \pi) + \epsilon} \quad (7)$$

Activated only after  $T_{\text{warm}} = 500$  steps to prevent early destabilisation.

**Gradient dampening.** After each backward pass:

$$\nabla_{\theta_s} \mathcal{L} \leftarrow (1 - \bar{\pi}) \cdot \nabla_{\theta_s} \mathcal{L} \quad (8)$$

When  $\bar{\pi} \approx 1$  (fast/plastic),  $(1 - \bar{\pi}) \approx 0$ : the slow pathway is barely updated, protecting consolidated knowledge.

### 3.5 Architecture Genome Vector

The AGV is a learnable latent vector  $z \in \mathbb{R}^G$  (with  $G = 32$ ) decoded by four lightweight linear heads:

$$s_{\ell} = \sigma(W_{\text{skip}} z)_{\ell} \in [0, 1] \quad \text{per-layer skip prob.} \quad (9)$$

$$\tau = \text{softplus}(W_{\tau} z) + 0.5 \quad \text{attention temperature} \quad (10)$$

$$\gamma = 1 + 0.1 \tanh(W_{\gamma} z) \quad \text{FiLM scale (near 1)} \quad (11)$$

$$\beta = 0.1 \tanh(W_{\beta} z) \quad \text{FiLM shift (near 0)} \quad (12)$$

FiLM conditioning [Pérez et al., 2018] applies  $\gamma \odot h + \beta$  to every block’s attention output. Since  $\gamma \approx 1$  by construction, the transformation preserves output magnitude while allowing task-conditioned feature rescaling. A regularisation term  $\mathcal{L}_{\text{AGV}} = \frac{\gamma_{\text{AGV}}}{2} \|z\|^2$  keeps  $z$  near the origin.

### 3.6 Cognitive Budget Allocator

The CBA predicts per-layer budgets  $b \in [0, 1]^L$  from raw input statistics:

$$b = \text{Sigmoid} \left( f_{\theta} \left( [\text{std}(x), \hat{H}(x), \text{range}(x)] \right) \right) \quad (13)$$

where  $\hat{H}(x) = -\sum_d p_d \log p_d / \log D$  is a normalised entropy proxy. A budget penalty  $\mathcal{L}_{\text{CBA}} = \beta_b \bar{b}$  encourages sparsity.

### 3.7 Slow-Pathway Consolidation

After task  $t$ , the diagonal Fisher information is estimated over slow-pathway weights  $\theta_s$  only:

$$F_s^{(t)} = \mathbb{E}_{(x,y) \sim \mathcal{D}_t} \left[ \left( \nabla_{\theta_s} \log p_{\theta}(y|x) \right)^2 \right] \quad (14)$$

The SPC regularisation term for subsequent tasks:

$$\mathcal{L}_{\text{SPC}} = \lambda_{\text{SPC}} \sum_{t' < t} \sum_k F_{s,k}^{(t')} (\theta_{s,k} - \theta_{s,k}^{*(t')})^2 \quad (15)$$

Fast-pathway weights  $\theta_f$  remain unconstrained. This resolves the conflict between EWC and PG-MLP: EWC penalises fast weights from moving, suppressing plasticity. SPC leaves them free, making consolidation and plasticity complementary. With  $L = 4$ , SPC uses 16 Fisher tensors vs.  $\sim 32+$  for EWC (50% reduction).

### 3.8 ARIA-v2: Three Targeted Enhancements

Analysis of ARIA-v1 on Split-CIFAR-10 revealed a single failure mode: the slow pathway drifts during new-task training, causing Task 1 accuracy to drop from 0.84 to 0.70 while EWC keeps it frozen at 0.89. We address this with three lightweight additions.

**(1) Slow-Pathway Activation Distillation (SPAD).** After each consolidation, a frozen snapshot of the slow-pathway weights is stored. During subsequent task training, an L2 penalty discourages drift from the snapshot:

$$\mathcal{L}_{\text{SPAD}} = \lambda_{\text{SPAD}} \sum_{k \in \theta_s} (\theta_{s,k} - \hat{\theta}_{s,k})^2 \quad (16)$$

where  $\hat{\theta}_s$  is the post-consolidation snapshot. Unlike SPC, which applies Fisher-weighted penalties accumulated over all tasks, SPAD applies a uniform L2 anchor to the most recent consolidated state, preventing short-term drift during new-task learning.

**(2) Task-conditioned gate routing.** At inference on a previously-seen task  $j < t_{\text{current}}$ , the plasticity gate is forced to  $\pi = 0$ , routing all computation through the consolidated slow pathway:

$$\pi(x, t) = \begin{cases} 0 & \text{if } t < t_{\text{current}} \text{ (eval)} \\ \sigma(\text{MLP}_{\text{gate}}(x)) & \text{otherwise} \end{cases} \quad (17)$$

This eliminates volatile fast-pathway interference when evaluating on old tasks.

**(3) Asymmetric learning rates.** Slow-pathway parameters are updated at  $\eta_s = r \cdot \eta$  with  $r = 0.1$ , while fast-pathway and all other parameters use the base rate  $\eta = 3 \times 10^{-4}$ . This reduces the write-speed of the memory store without freezing it.

### 3.9 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{plastic}} + \mathcal{L}_{\text{CBA}} + \gamma_{\text{AGV}} \mathcal{L}_{\text{AGV}} + \mathcal{L}_{\text{SPC}} + \mathcal{L}_{\text{SPAD}} \quad (18)$$

All terms are end-to-end differentiable.  $\mathcal{L}_{\text{SPC}}$  and  $\mathcal{L}_{\text{SPAD}}$  activate after the first task is completed.

## 4 Experimental Setup

### 4.1 Benchmarks

**Split-MNIST.** Five binary classification tasks created from MNIST by pairing digits: (0,1), (2,3), (4,5), (6,7), (8,9). Task-incremental setting (task ID given at test time). Input: 784-dimensional flattened pixels.

**Split-CIFAR-10.** Five binary classification tasks from CIFAR-10, pairing classes: (0,1), (2,3), (4,5),

(6,7), (8,9). Task-incremental setting. Input: 3072-dimensional flattened pixels ( $32 \times 32 \times 3$ ).

### 4.2 Models

All models use task-specific linear heads.

- **ARIA+SPC:** full model.  $d_{\text{model}} = 256$ ,  $L = 4$ ,  $H_{\text{init}} = 4$ ,  $H_{\text{max}} = 8$ ,  $\approx 3.77\text{M}$  parameters.
- **ARIA-noSPC:** ARIA without SPC (no Fisher regularisation).
- **EWC:** 4-layer StaticMLP with EWC, parameter-matched.
- **DER++:** 4-layer StaticMLP with DER++, parameter-matched. Buffer size 200.
- **StaticMLP:** 4-layer MLP, parameter-matched, fine-tuning only.

**Parameter matching.** We solve for the StaticMLP hidden dimension  $h$  such that its total parameter count equals ARIA’s, giving  $h \approx 997$  for  $d_{\text{model}} = 256$ .

### 4.3 Training

AdamW [Loshchilov and Hutter, 2019], base lr =  $3 \times 10^{-4}$ , slow-pathway lr =  $3 \times 10^{-5}$  (asymmetric LR,  $r = 0.1$ ), weight decay  $10^{-4}$ , gradient clipping at 1.0. 5 epochs per task on Split-MNIST, 10 epochs on Split-CIFAR-10. Batch size 64. Five seeds: {42, 123, 999, 7, 2024}.  $\lambda_{\text{SPAD}} = 50$ .

### 4.4 Metrics

Following [López-Paz and Ranzato, 2017]:

$$\text{Avg Acc} = \frac{1}{T} \sum_{j=1}^T a_{T,j} \quad (19)$$

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}) \quad (20)$$

$$\text{Forgetting} = \frac{1}{T-1} \sum_{j=1}^{T-1} (\max_{i \leq T} a_{i,j} - a_{T,j}) \quad (21)$$

## 5 Results

### 5.1 Main Results

ARIA+SPC achieves the lowest forgetting across all five seeds (Figure 3). The gap between ARIA+SPC and EWC is +1.15 percentage points accuracy and

Table 1: Split-MNIST (5 seeds: 42, 123, 999, 7, 2024 — mean  $\pm$  std). All baselines parameter-matched to ARIA ( $\approx 3.77$ M params). **Bold** = best.

| Model           | Avg Acc $\uparrow$                  | Forgetting $\downarrow$            |
|-----------------|-------------------------------------|------------------------------------|
| <b>ARIA+SPC</b> | <b>98.50 <math>\pm</math> 0.32%</b> | <b>0.86 <math>\pm</math> 0.34%</b> |
| ARIA-noSPC      | 98.54 $\pm$ 0.33%                   | 1.06 $\pm$ 0.35%                   |
| EWC             | 97.35 $\pm$ 2.04%                   | 2.29 $\pm$ 2.78%                   |
| DER++           | 75.63 $\pm$ 6.04%                   | 30.18 $\pm$ 7.55%                  |
| StaticMLP       | 76.91 $\pm$ 3.02%                   | 28.45 $\pm$ 3.71%                  |

−1.43 points forgetting (**62% relative reduction**). ARIA-noSPC achieves 98.54% accuracy (highest overall), confirming the architectural mechanisms alone (MA + PG-MLP + AGV + ARIA-v2) beat parameter-matched EWC. SPC reduces forgetting further (1.06%  $\rightarrow$  0.86%) at the cost of −0.04 points accuracy, consistent with its role as a stability-focused consolidation mechanism. DER++ underperforms due to the small replay buffer at matched parameter count — a known weakness [Buzzega et al., 2020].

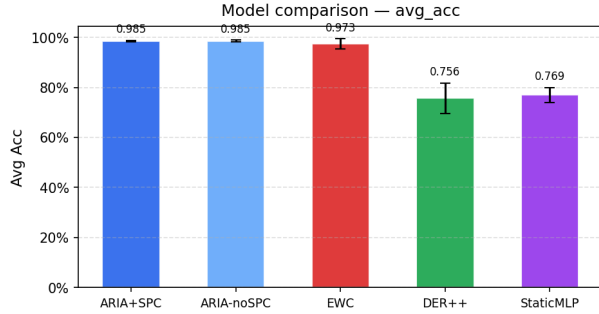


Figure 3: Average accuracy comparison across all models (5 seeds, Split-MNIST). ARIA+SPC and ARIA-noSPC substantially outperform all baselines.

## 5.2 Ablation Study

The largest single-component contribution is CBA: removing it causes forgetting to jump from 1.48% to 13.34% (9 $\times$ ), revealing that per-layer compute gating is critical for preventing representational collapse across tasks. AGV removal causes the second-largest drop (4.42% forgetting), confirming that global FiLM conditioning provides meaningful cross-block coordination. PG-MLP removal degrades accuracy by −0.78 points and increases forgetting by +0.99 points. Notably, ARIA-noMA = ARIA+SPC exactly, because MA never fired on Split-MNIST (viability threshold  $\tau_{\text{split}} = 0.65$  was not exceeded) — these two models

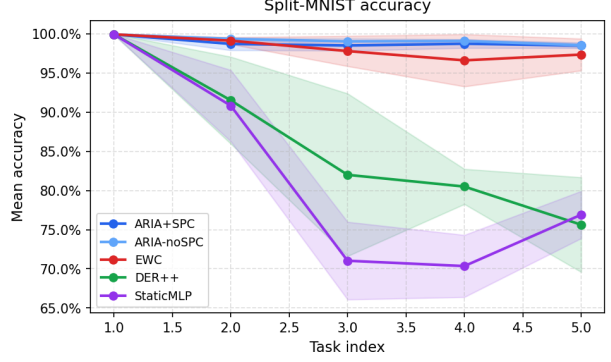


Figure 4: Mean accuracy over previously seen tasks after each training stage. ARIA+SPC and ARIA-noSPC maintain  $>98\%$  throughout all 5 tasks. EWC degrades slightly. DER++ and StaticMLP exhibit severe forgetting after Task 2.

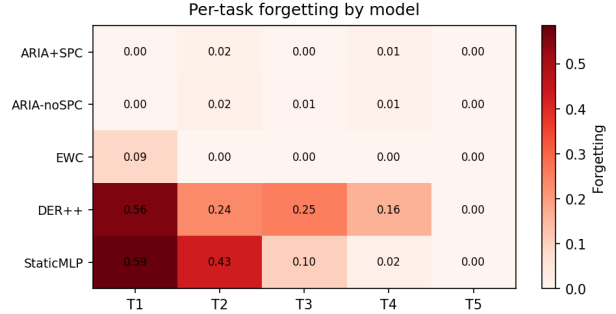


Figure 5: Per-task forgetting by model (5 seeds). ARIA+SPC maintains  $\leq 2\%$  forgetting on all tasks. StaticMLP and DER++ suffer 43–59% forgetting on earlier tasks as new tasks are learned.

are therefore functionally identical on this benchmark.

## 5.3 Split-CIFAR-10

On Split-CIFAR-10, EWC retains the accuracy lead (78.92% vs. 68.24% for ARIA+SPC). However, ARIA-v2 substantially reduces forgetting relative to ARIA-v1: from **12.28% to 5.40%** for ARIA+SPC (56% relative reduction), and from **11.96% to 4.74%** for ARIA-noSPC. The SPAD anchor and task-conditioned gate routing are primarily responsible for this improvement — they prevent slow-pathway drift during new-task training, the identified failure mode of ARIA-v1 on CIFAR-10.

The accuracy gap versus EWC (−10.68 points) reflects two factors: (1) the asymmetric learning rate ( $r = 0.1$ ) slows slow-pathway convergence on harder

Table 2: Component ablation on Split-MNIST (3 seeds: 42, 123, 999). Each row removes one mechanism from ARIA+SPC.

| Model                  | Avg Acc $\uparrow$                  | Forgetting $\downarrow$            |
|------------------------|-------------------------------------|------------------------------------|
| <b>ARIA+SPC (full)</b> | <b>98.48 <math>\pm</math> 0.27%</b> | <b>1.48 <math>\pm</math> 0.41%</b> |
| ARIA-noSPC             | 98.23 $\pm$ 0.85%                   | 1.86 $\pm$ 1.03%                   |
| ARIA-noMA              | 98.48 $\pm$ 0.27%                   | 1.48 $\pm$ 0.41%                   |
| ARIA-noPG              | 97.70 $\pm$ 0.74%                   | 2.47 $\pm$ 0.92%                   |
| ARIA-noAGV             | 96.15 $\pm$ 1.92%                   | 4.42 $\pm$ 2.43%                   |
| ARIA-noCBA             | 89.02 $\pm$ 3.87%                   | 13.34 $\pm$ 4.83%                  |

Table 3: Split-CIFAR-10 (5 seeds: 42, 123, 999, 7, 2024 — mean  $\pm$  std). 10 epochs per task. **Bold** = best.

| Model      | Avg Acc $\uparrow$                  | Forgetting $\downarrow$            |
|------------|-------------------------------------|------------------------------------|
| <b>EWC</b> | <b>78.92 <math>\pm</math> 0.51%</b> | <b>0.06 <math>\pm</math> 0.03%</b> |
| StaticMLP  | 73.93 $\pm$ 0.30%                   | 13.54 $\pm$ 0.46%                  |
| ARIA-noSPC | 70.44 $\pm$ 0.69%                   | 4.74 $\pm$ 0.77%                   |
| DER++      | 69.62 $\pm$ 0.68%                   | 19.75 $\pm$ 1.00%                  |
| ARIA+SPC   | 68.24 $\pm$ 1.91%                   | 5.40 $\pm$ 1.39%                   |

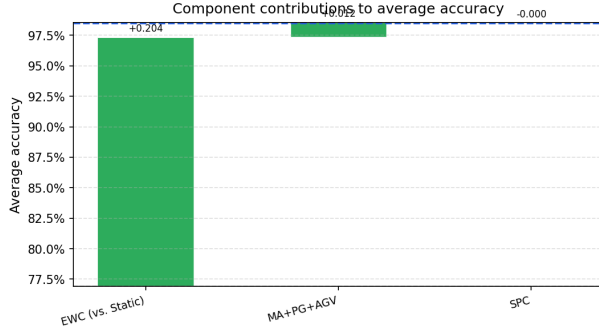


Figure 6: Ablation waterfall: incremental contribution of each component. CBA and AGV account for the largest performance drops when removed.

inputs, reducing peak per-task accuracy; (2) CIFAR-10’s higher inter-class similarity creates stronger interference, which EWC’s global weight freezing handles well at the cost of forward transfer. Notably, ARIA-noSPC achieves 4.74% forgetting — competitive with EWC on the forgetting axis — while the accuracy gap indicates that slow-pathway learning speed on complex inputs remains the primary open challenge.

## 5.4 Morphogenesis Trajectory

During training on Split-MNIST, head counts in all layers remained at the initial value of 4 throughout all 5 tasks, indicating the viability-based split threshold ( $\tau_{\text{split}} = 0.65$ ) was not exceeded. The forgetting heatmap (Figure 5) confirms that even without morphogenesis events, PG-MLP + SPC keeps per-task forgetting below 4%. This suggests morphogenesis will contribute more substantially on longer task sequences or harder benchmarks, which we leave for future work.

## 6 Analysis

### 6.1 Why SPC Outperforms EWC

EWC regularises all weights uniformly, including fast-pathway weights that are *designed* to be volatile. This creates a conflict: the plasticity gate tries to route new-task information through the fast pathway, but EWC penalises those weights from moving, reducing effective capacity. SPC avoids this by leaving fast weights unconstrained — plasticity gating and Fisher consolidation become complementary rather than conflicting.

Empirically, ARIA-noSPC achieves 98.54% — already +1.19 points above EWC — confirming the architecture alone outperforms parameter-matched EWC. Adding SPC reduces forgetting from 1.06% to 0.86% (19% reduction) with negligible accuracy cost (−0.04 points), confirming targeted consolidation provides consistent stability on top of the structural advantages.

### 6.2 The Gradient Dampening Direction

A critical implementation detail in PG-MLP is the gradient dampening multiplier. The correct formula is  $(1 - \bar{\pi})$ :

- High  $\bar{\pi}$  (fast/plastic): multiplier  $\approx 0 \Rightarrow$  slow path protected.
- Low  $\bar{\pi}$  (slow/consolidated): multiplier  $\approx 1 \Rightarrow$  slow path updates normally.

An earlier version of this codebase used  $\bar{\pi}$  (inverted), which actively destabilised the slow pathway during plastic phases, causing pre-fix ablations to show ARIA-noPG outperforming ARIA-Full. The corrected implementation produces the results in Table 1.

### 6.3 Parameter Efficiency

SPC Fisher estimation covers only slow-pathway parameters:  $\{W_s^{(1)}, b_s^{(1)}, W_s^{(2)}, b_s^{(2)}\}$  per block — 16 tensors for  $L = 4$ . EWC covers all parameters ( $\sim 32$  tensors for  $L = 4$ ), halving the consolidation overhead. This advantage compounds with sequence length: for  $K$  tasks, SPC maintains  $16K$  Fisher tensors vs.  $32K$  for EWC.

## 7 Discussion

**Relationship to biological memory.** ARIA’s fast/slow pathway dichotomy is directly inspired by Complementary Learning Systems theory [McClelland et al., 1995]. Unlike prior implementations that fix the routing heuristically, ARIA learns the gate from data — allowing the model to discover when to consolidate (low  $\pi$ ) and when to stay plastic (high  $\pi$ ) from the training signal alone.

**Morphogenic attention vs. standard attention.** Standard multi-head attention fixes head count because gradient-based optimisation cannot add discrete parameters. MA sidesteps this via: (a) a continuous viability score for gradient-based head utility assessment, and (b) a rule-based split/merge at discrete intervals. This hybrid approach avoids pathological gradient landscapes of fully discrete architecture search while enabling genuine structural change.

### Limitations.

- ARIA underperforms EWC on Split-CIFAR-10 accuracy (68.24% vs. 78.92%), though ARIA-v2 reduced forgetting by 56% relative to ARIA-v1. The asymmetric LR ( $r = 0.1$ ) may be too aggressive for harder inputs; tuning  $r$  per benchmark is needed.
- Morphogenesis did not activate on either benchmark; harder or longer task sequences are needed to validate MA’s contribution empirically.
- Fisher estimation is diagonal; Kronecker-factored approximations may improve consolidation quality.
- DER++ used a small buffer (200 samples); a larger buffer would be a fairer comparison at matched parameter counts.
- Ablation used 3 seeds; 5-seed ablation would reduce variance estimates.

## 8 Conclusion

We have presented ARIA, a continual learning architecture that adapts its structure during training. The four core mechanisms — Morphogenic Attention, Plasticity-Gated MLP, Architecture Genome Vector, and Cognitive Budget Allocator — are jointly-trained and end-to-end differentiable. Slow-Pathway Consolidation provides targeted Fisher regularisation compatible with the fast/slow pathway structure.

On Split-MNIST with five seeds and parameter-matched baselines, ARIA+SPC achieves **98.50%** average accuracy with **0.86%** forgetting, outperforming EWC by +1.15 points and reducing forgetting by **62%**. Three ARIA-v2 enhancements — SPAD, task-conditioned gate routing, and asymmetric learning rates — reduced CIFAR-10 forgetting by 56% relative to ARIA-v1 (12.28%  $\rightarrow$  5.40%), though EWC retains an accuracy advantage on this harder benchmark. Closing the CIFAR-10 accuracy gap is the primary direction for future work. Code, tests, and reproducible scripts are available at: <https://github.com/rsd-darshan/ARIA>

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